

How about more categories?

One-hot

- There are a few ways of modeling predictors with many categories:
 - **Contrasts**: Each category is compared to a **baseline**, and the coefficients are comparisons between baseline and levels
 - **One-hot**: coefficients are means of each level of the predictor
 - **Residuals**: an overall mean is measured, and coefficients are differences between each level and the global mean

```
> x
[1] "B" "A" "A" "B" "C" "B" "B" "C" "B"
> onehot
  xA  xB  xC
1  0  1  0
2  1  0  0
3  1  0  0
4  0  1  0
5  0  0  1
6  0  1  0
7  0  1  0
8  0  0  1
9  0  1  0
```

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```
> onehot = model.matrix(~0+x, data = df)
> fit_onehot = stan_glm(y ~ 0 + onehot, data =
df, cores = 4)
> summary(fit_onehot)[1:3, 1:3]
```

	mean	mcse	sd
onehotxA	0.72	0	0.13
onehotxB	2.04	0	0.14
onehotxC	2.93	0	0.12